# AI and Theorem Proving

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New Technologies in Mathematics Seminar
January 13, 2021, Harvard University

European Research Council
Established by the European Commission

# Outline

Motivation, Learning vs. Reasoning

Computer Understandable (Formal) Math

Learning of Theorem Proving

Examples and Demos

High-level Reasoning Guidance: Premise Selection

Low Level Guidance of Theorem Provers

Mid-level Reasoning Guidance

More on Neural Guidance, Synthesis and Conjecturing

Autoformalization

# How Do We Automate Math and Science?

- What is mathematical and scientific thinking?
- Pattern-matching, analogy, induction from examples
- Deductive reasoning
- Complicated feedback loops between induction and deduction
- Using a lot of previous knowledge - both for induction and deduction

- We need to develop such methods on computers
- Are there any large corpora suitable for nontrivial deduction?
- Yes! Large libraries of formal proofs and theories
- So let's develop strong AI on them!

# What is Formal Mathematics?

- Developed thanks to the Leibniz/Russell/Frege/Hilbert/... program
- Mathematics put on formal logic foundations (*symbolic computation*)
- ... which btw. led also to the rise of computers (Turing/Church, 1930s)
- Formal math (1950/60s): combine formal foundations and the newly available computers
- De Bruijn, Milner, Trybulec, Boyer and Moore, Gordon, Huet, Paulson, ...
- Automath, LCF, Mizar, NQTHM and ACL2, HOL, Coq, Isabelle, ...
- Conceptually very simple:
- Write all your axioms and theorems so that computer understands them
- Write all your inference rules so that computer understands them
- Use the computer to check that your proofs follow the rules
- But in practice, it turns out not to be so simple
- Many approaches, still not mainstream, but big breakthroughs recently

# History and Motivation for AI/TP

- Intuition vs Formal Reasoning – Poincaré vs Hilbert, Science & Method
- Turing's 1950 paper: Learning Machines, learn Chess?, undecidability??
- Lenat, Langley, etc: manually-written heuristics, learn Kepler laws,...
- Denzinger, Schulz, Goller, Fuchs – late 90's, ATP-focused:
- *Learning from Previous Proof Experience*
- My MSc (1998): Try ILP to learn rules and heuristics from IMPS/Mizar
- Since: Use large formal math (Big Proof) corpora: Mizar, Isabelle, HOL
- ... to combine/develop symbolic/statistical deductive/inductive ML/TP/AI
- ... hammer-style methods, feedback loops, internal guidance, ...
- More details – AGI'18 keynote: `https://bit.ly/3qifhg4`
- AI vs DL: Ben Goertzel's Prague talk: `https://youtu.be/Zt2HSTuGBn8`
- Big AI visions: automate/verify math, science, law, (Leibniz, McCarthy, ..)
- Practical impact: boost today's large ITP verification projects

# Using Learning to Guide Theorem Proving

- **high-level**: pre-select lemmas from a large library, give them to ATPs
- **high-level**: pre-select a good ATP strategy/portfolio for a problem
- **high-level**: pre-select good *hints* for a problem, use them to guide ATPs
- **low-level**: guide every inference step of ATPs (tableau, superposition)
- **low-level**: guide every kernel step of LCF-style ITPs
- **mid-level**: guide application of tactics in ITPs
- **mid-level**: invent suitable ATP strategies for classes of problems
- **mid-level**: invent suitable conjectures for a problem
- **mid-level**: invent suitable concepts/models for problems/theories
- **proof sketches**: explore stronger/related theories to get proof ideas
- **theory exploration**: develop interesting theories by conjecturing/proving
- **feedback loops**: (dis)prove, learn from it, (dis)prove more, learn more, ...
- **autoformalization**: (semi-)automate translation from LaTeX to formal
- ...

# Large AI/TP Datasets

- Mizar / MML / MPTP – since 2003
- MPTP Challenge (2006), MPTP2078 (2011), Mizar40 (2013)
- Isabelle (and AFP) – since 2005
- Flyspeck (including core HOL Light and Multivariate) – since 2012
- HOL4 – since 2014, CakeML – 2017, GRUNGE – 2019
- Coq – since 2013/2016
- ACL2 – 2014?
- Lean?, Stacks?, Arxiv?, ProofWiki?, ...

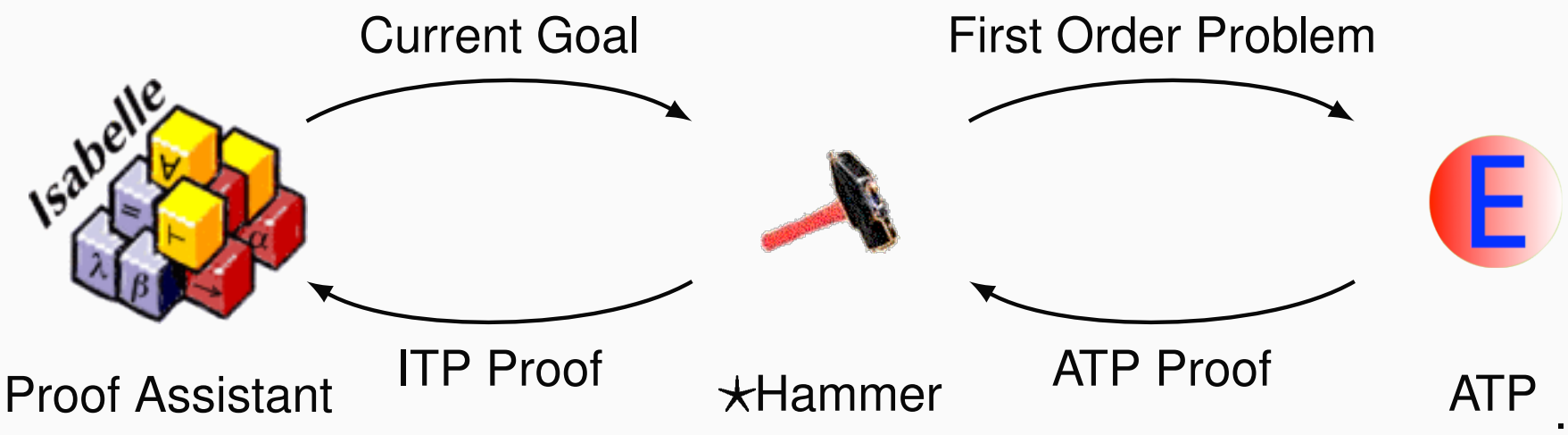
# Demos

- ENIGMA/hammer proofs of Pythagoras : `https://bit.ly/2MVPAn7` (more at `http://grid01.ciirc.cvut.cz/~mptp/enigma-ex.pdf`) and simplified Carmichael `https://bit.ly/3oGBdRz`
- Hammering demo: `http://grid01.ciirc.cvut.cz/~mptp/out4.ogv`
- TacticToe on HOL4: `http://grid01.ciirc.cvut.cz/~mptp/tactictoe_demo.ogv`
- Tactician for Coq: `https://blaauwbroek.eu/papers/cicm2020/demo.mp4`, `https://coq-tactician.github.io/demo.html`
- Inf2formal over HOL Light: `http://grid01.ciirc.cvut.cz/~mptp/demo.ogv`

# High-level ATP guidance: Premise Selection

- Early 2003: Can existing ATPs be used over the freshly translated Mizar library?
- About 80000 nontrivial math facts at that time – impossible to use them all
- Is good premise selection for proving a new conjecture possible at all?
- Or is it a mysterious power of mathematicians? (Penrose)
- Today: Premise selection is not a mysterious property of mathematicians!
- Reasonably good algorithms started to appear (more below).
- Extensive human (math) knowledge obsolete?? (cf. Watson, Debater, ..)
- Since 2004 (my PhD): many examples of nontrivial alternative proofs proposed by the AIs - in Mizar, Flyspeck, Isabelle, ..
- The premise selection algorithms see *wider* than humans

# Today's AI-ATP systems ($\star$-Hammers)



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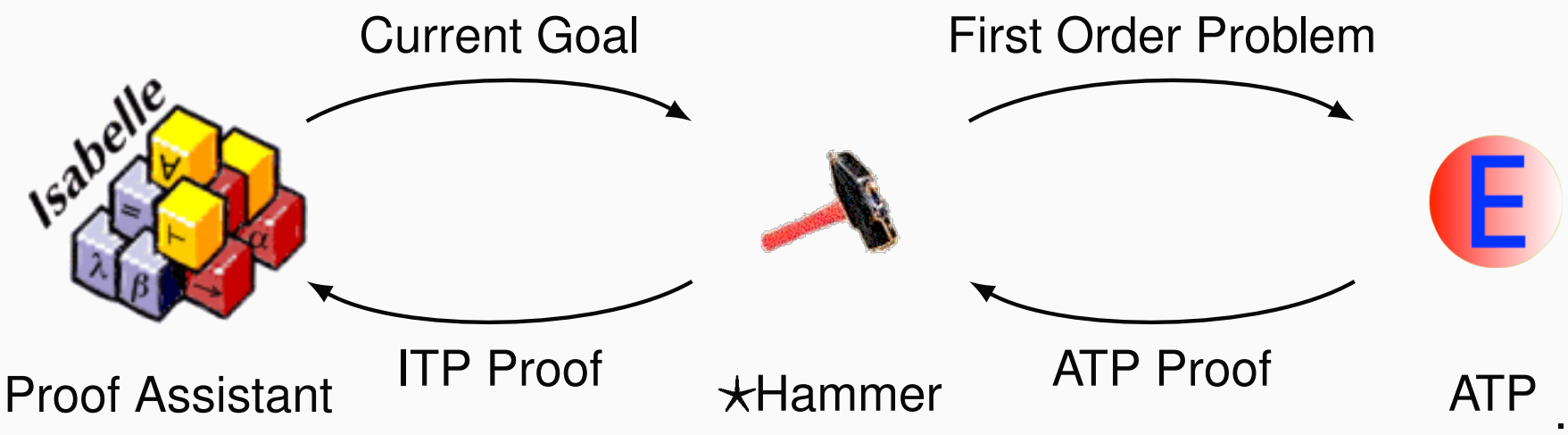


How much can it do?

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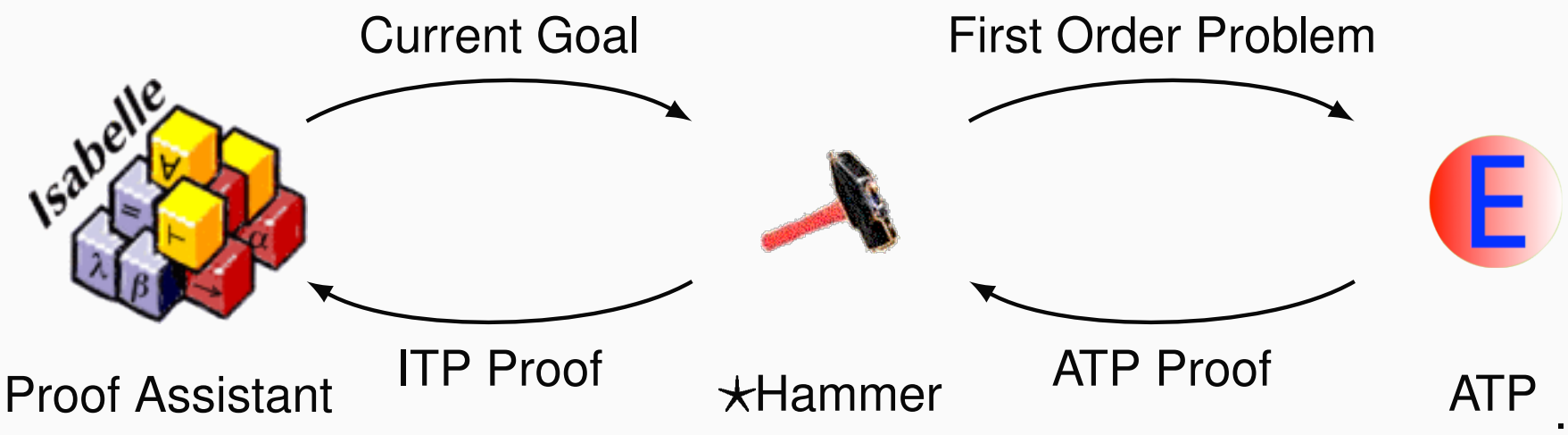


How much can it do?

- Mizar / MML – MizAR
- Isabelle (Auth, Jinja) – Sledgehammer
- Flyspeck (including core HOL Light and Multivariate) – HOL(y)Hammer
- HOL4 (Gauthier and Kaliszyk)
- CoqHammer (Czajka and Kaliszyk) - about 40% on Coq standard library

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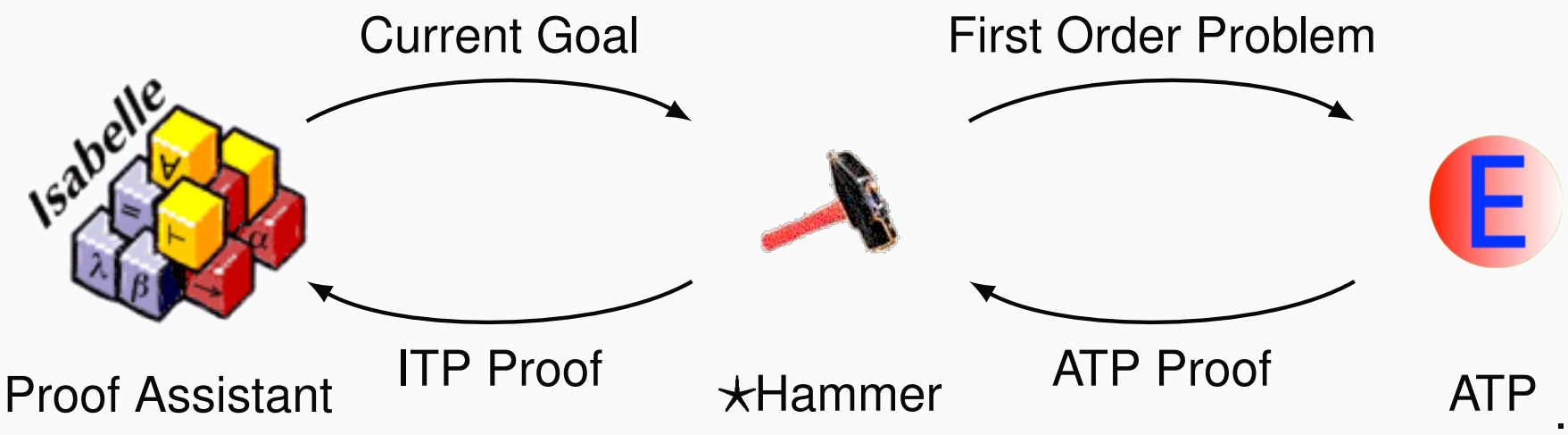


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$\approx$ 40-45% success rate (close to 60% on Mizar as of 2021)

# Premise Selection and Hammer Methods

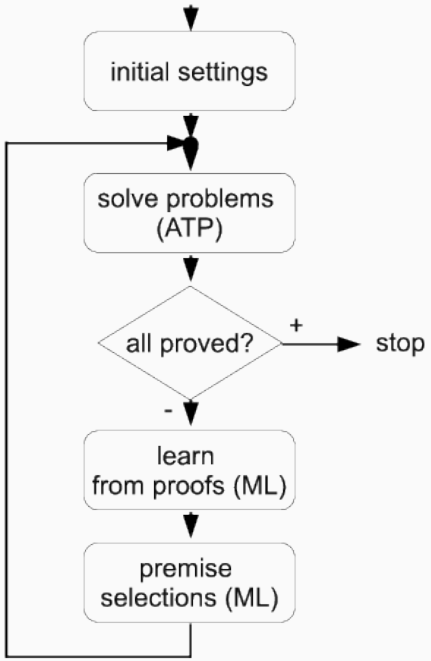
- Many syntactic features (symbols, walks in the parse trees)
- More semantic features encoding
- term matching/unification, validity in models, latent semantics (LSI)
- Distance-weighted k-nearest neighbor, SVMs, Naive Bayes
- Gradient boosted decision trees (GBDTs - XGBoost, LightGBM)
- Neural models: CNNs, RNNs/Attention/Transformers/GPT, GNNs
- As of 2020, tough competition between GBDTs, GNNs and RNNs/Transformers (and relatives)
- K-NN still very good, Olsak's logic-aware GNN probably best
- RNNs/Transformers good at stateful premise selection (Piotrowski 2019,2020)
- Ensemble methods combining the different predictors help a lot

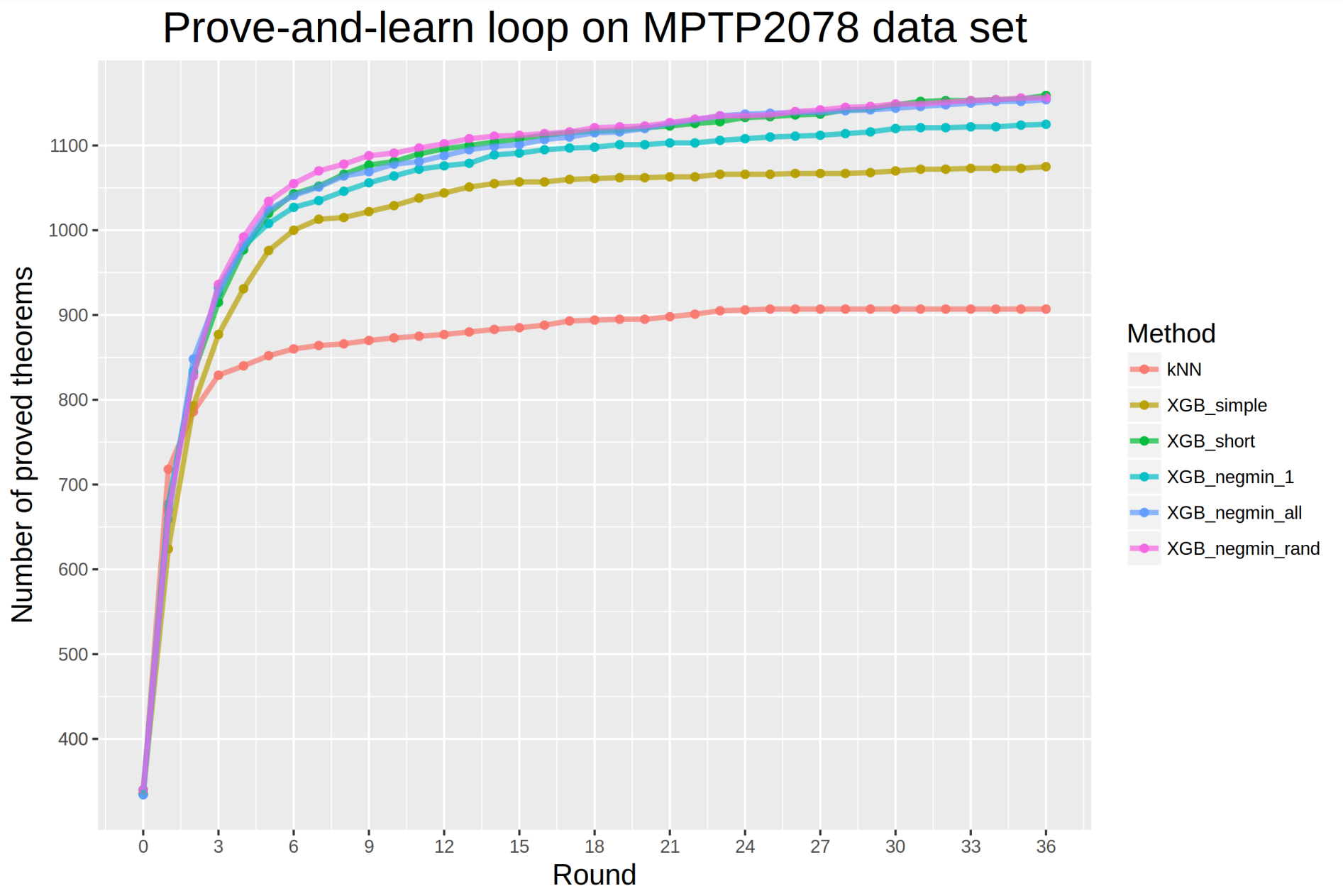
# Premise Selection and Hammer Methods

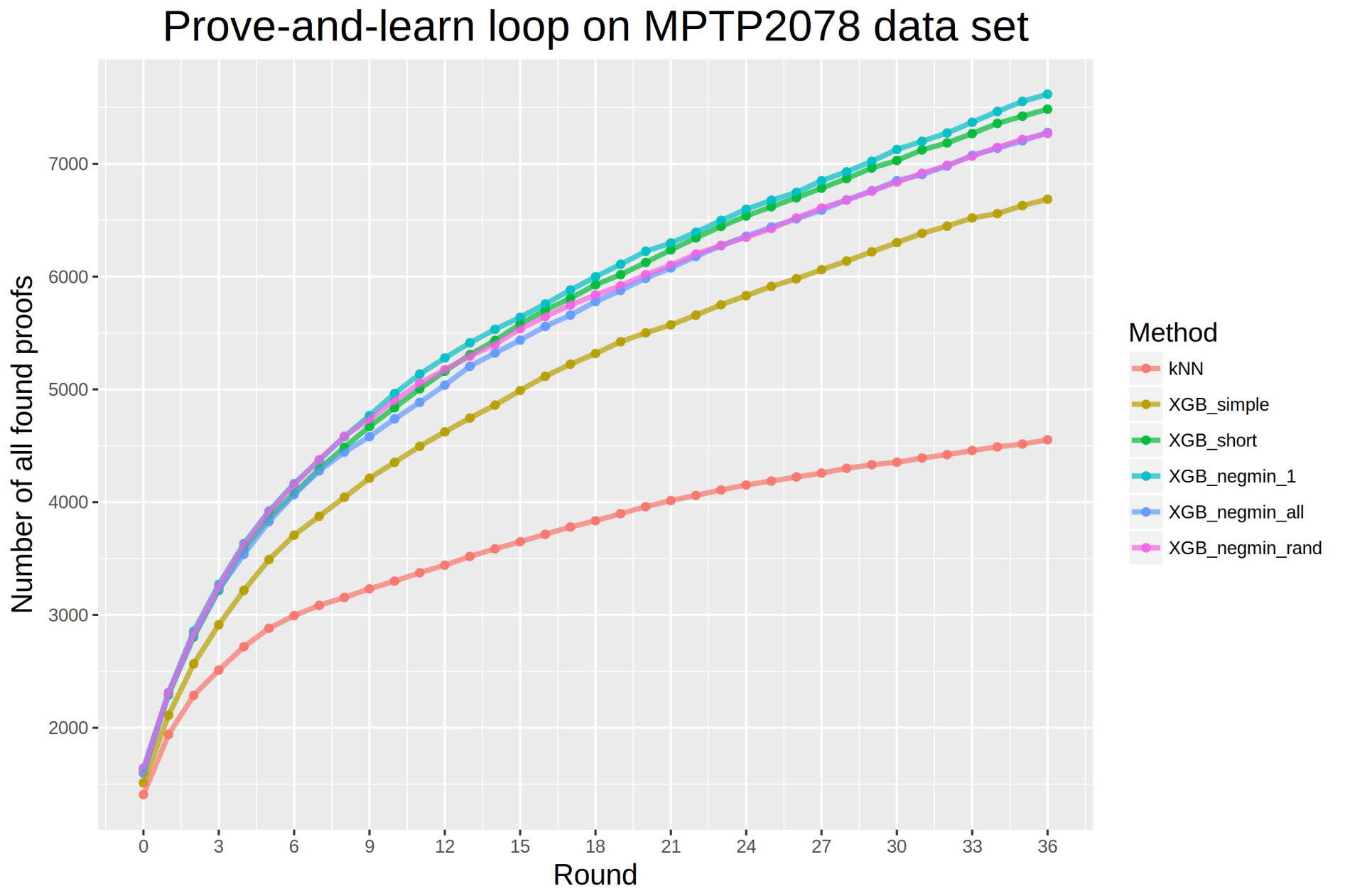
- Learning in a binary setting from many alternative proofs
- Interleaving many learning and proving runs (*MaLARea loop*) to get positives/negatives (ATPBoost - Piotrowski 2018)
- Matching and transferring concepts and theorems between libraries (Gauthier & Kaliszyk) – allows “superhammers”, conjecturing, and more
- Lemmatization – extracting and considering millions of low-level lemmas and learning from their proofs
- Hammers combined with guided tactical search: TacticToe (Gauthier - HOL4) and its later relatives

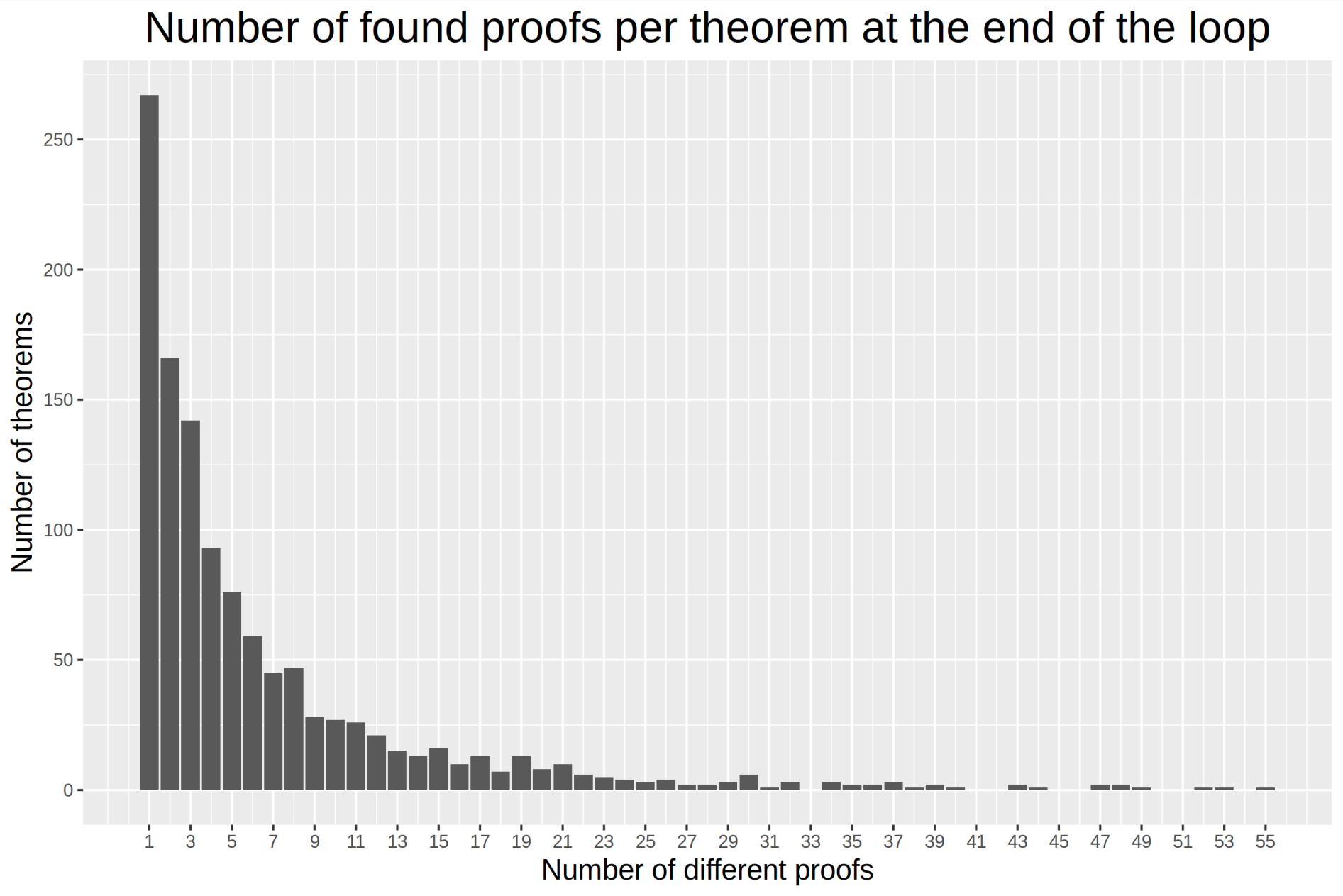
# High-level feedback loops – MALARea, ATPBoost

- Machine Learner for Autom. Reasoning (2006) – infinite hammering
- feedback loop interleaving ATP with learning premise selection
- both syntactic and semantic features for characterizing formulas:
- evolving set of finite (counter)models in which formulas evaluated
- winning AI/ATP benchmarks (MPTPChallenge, CASC 2008/12/13/18)
- ATPBoost (Piotrowski) - recent incarnation focusing on multiple proofs




Prove-and-learn loop on MPTP2078 data set
Number of proved theorems
1100
1000
900
800
700
600
500
400
0
3
6
9
12
15
18
21
24
27
30
33
36
Round
Method
kNN
XGB_simple
XGB_short
XGB_negmin_1
XGB_negmin_all
XGB_negmin_rand


Prove-and-learn loop on MPTP2078 data set
Number of all found proofs
7000
6000
5000
4000
3000
2000
0
3
6
9
12
15
18
21
24
27
30
33
36
Round
Method
kNN
XGB_simple
XGB_short
XGB_negmin_1
XGB_negmin_all
XGB_negmin_rand


Number of found proofs per theorem at the end of the loop
Number of theorems
0
50
100
150
200
250
1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49 51 53 55
Number of different proofs

# Low-level: Statistical Guidance of Connection Tableau

- learn guidance of every clausal inference in connection tableau (leanCoP)
- set of first-order clauses, *extension* and *reduction* steps
- proof finished when all branches are closed
- a lot of nondeterminism, requires backtracking
- *Iterative deepening* used in leanCoP to ensure completeness
- good for learning – the tableau compactly represents the proof state

Clauses:

$c_1 : P(x)$

$c_2 : R(x, y) \vee \neg P(x) \vee Q(y)$

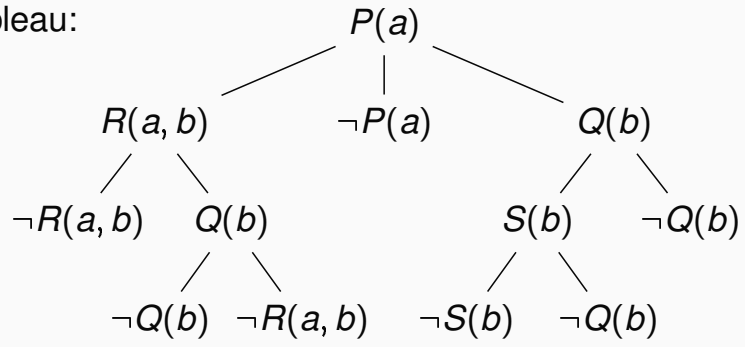
$c_3 : S(x) \vee \neg Q(b)$

$c_4 : \neg S(x) \vee \neg Q(x)$

$c_5 : \neg Q(x) \vee \neg R(a, x)$

$c_6 : \neg R(a, x) \vee Q(x)$

Closed Connection Tableau:



# Statistical Guidance of Connection Tableau

- **MaLeCoP** (2011): first prototype Machine Learning Connection Prover
- extension rules chosen by naive Bayes trained on good decisions
- training examples: tableau features plus the name of the chosen clause
- initially slow: off-the-shelf learner 1000 times slower than raw leanCoP
- 20-time search shortening on the MPTP Challenge
- second version: 2015, with C. Kaliszyk
- both prover and naive Bayes in OCAML, fast indexing
- Fairly Efficient MaLeCoP = **FEMaLeCoP**
- 15% improvement over untrained leanCoP on the MPTP2078 problems
- using iterative deepening - enumerate shorter proofs before longer ones

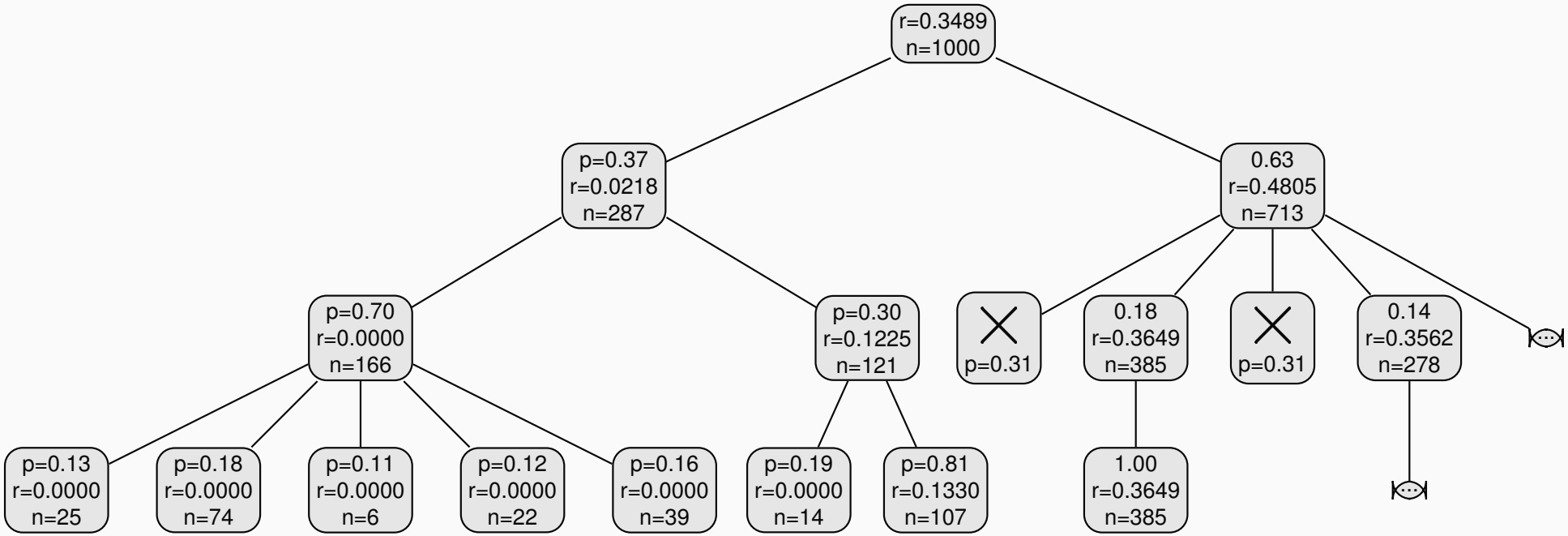
# Statistical Guidance of Connection Tableau – rlCoP

- 2018: stronger learners via C interface to OCAML (boosted trees)
- remove iterative deepening, the prover can go arbitrarily deep
- added Monte-Carlo Tree Search (MCTS)
- MCTS search nodes are sequences of clause application
- a good heuristic to explore new vs exploit good nodes:

$$\frac{w_i}{n_i} + c \cdot p_i \cdot \sqrt{\frac{\ln N}{n_i}} \qquad \text{(UCT - Kocsis, Szepesvari 2006)}$$

- learning both *policy* (clause selection) and *value* (state evaluation)
- clauses represented not by names but also by features (generalize!)
- binary learning setting used: | proof state | clause features |
- mostly term walks of length 3 (trigrams), hashed into small integers
- many iterations of proving and learning

# Tree Example



# Statistical Guidance of Connection Tableau – rlCoP

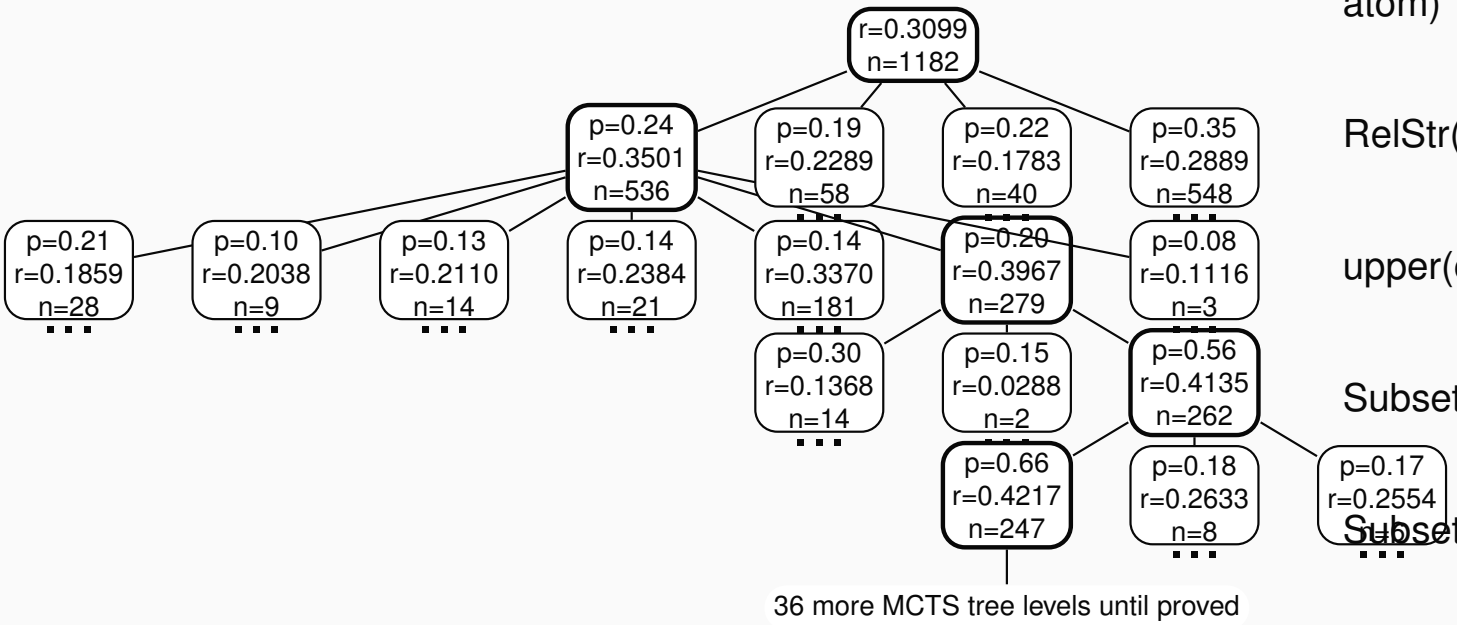
- On 32k Mizar40 problems using 200k inference limit
- nonlearning CoPs:

| System | leanCoP | bare prover | rlCoP no policy/value (UCT only) |
|---|---|---|---|
| Training problems proved | 10438 | 4184 | 7348 |
| Testing problems proved | **1143** | 431 | 804 |
| Total problems proved | 11581 | 4615 | 8152 |

- rlCoP with policy/value after 5 proving/learning iters on the training data
- $1624/1143 = 42.1\%$ improvement over leanCoP on the testing problems

| Iteration | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Training proved | 12325 | 13749 | 14155 | 14363 | 14403 | 14431 | 14342 | **14498** |
| Testing proved | 1354 | 1519 | 1566 | 1595 | **1624** | 1586 | 1582 | 1591 |

# More trees




# (tableau starting atom)

RelStr(c1)

upper(c1)

Subset(union(c2),carrier(c1))

Subset(c2,powerset(carrier(c1)

# Recent CoP Mutants: FLoP, GNN, RNN, lazyCoP

- FLoP – Finding Longer Proofs (Zombori et al, 2019)
- Curriculum Learning used for connection tableau over Robinson Arithmetic
- addition and multiplication learned perfectly from $1 * 1 = 1$
- headed towards learning algorithms/decision procedures from math data
- currently black-box, combinations with symbolic methods (ILP) our next target
- Using RNNs for better tableau encoding, prediction of actions ...
- ... even guessing (decoding) next tableau literals (Piotrowski 2020)
- plCoP (Zombori 20), GNN-CoP (Olsak 20), lazyCoP (Rawson)
- Zombori: learning new explainable Prolog actions (tactics) from proofs

# ENIGMA: Guiding the Best ATPs like E Prover

- harder for learning than tableau
- the proof state are two large heaps of clauses *processed*/*unprocessed*
- 2017: ENIGMA - manual feature engineering (Jakubuv & JU 2017)
- 2017: Deep guidance (neural nets) (Loos et al. 2017)
- both learn on E's proof search traces, put classifier in E
- positive examples: given clauses used in the proof
- negative examples: given clauses not used in the proof

# ENIGMA: Guiding the Best ATPs like E Prover

- ENIGMA (Jan Jakubuv 2017)
- Fast/hashed feature extraction followed by fast/sparse linear classifier
- about 80% improvement on the AIM benchmark
- Deep guidance: convolutional nets - too slow to be competitive
- ENIGMA-NG: better features and ML, gradient-boosted trees, tree NNs
- NNs made competitive in real-time, boosted trees still best
- 2020: fast GNN added (Olsak, Jakubuv), now competitive with GBDTs
- However very different: the GNN scores many clauses (context and query) simultaneously in a large graph

# Feedback loop for ENIGMA on Mizar data

- Similar to rlCoP - interleave proving and learning of ENIGMA guidance
- Done on 57880 Mizar problems recently
- Serious ML-guidance breakthrough applied to the best ATPs
- Ultimately a 70% improvement over the original strategy in 2019
- From 14933 proofs to 25397 proofs (all 10s CPU - no cheating)
- Went up to 40k in more iterations and 60s time in 2020

| | $\mathcal{S}$ | $\mathcal{S} \odot \mathcal{M}_9^0$ | $\mathcal{S} \oplus \mathcal{M}_9^0$ | $\mathcal{S} \odot \mathcal{M}_9^1$ | $\mathcal{S} \oplus \mathcal{M}_9^1$ | $\mathcal{S} \odot \mathcal{M}_9^2$ | $\mathcal{S} \oplus \mathcal{M}_9^2$ | $\mathcal{S} \odot \mathcal{M}_9^3$ | $\mathcal{S} \oplus \mathcal{M}_9^3$ |
|---|---|---|---|---|---|---|---|---|---|
| solved | 14933 | 16574 | 20366 | 21564 | 22839 | 22413 | 23467 | 22910 | 23753 |
| $\mathcal{S}\%$ | +0% | +10.5% | +35.8% | +43.8% | +52.3% | +49.4% | +56.5% | +52.8% | +58.4 |
| $\mathcal{S}+$ | +0 | +4364 | +6215 | +7774 | +8414 | +8407 | +8964 | +8822 | +9274 |
| $\mathcal{S}-$ | -0 | -2723 | -782 | -1143 | -508 | -927 | -430 | -845 | -454 |

| | $\mathcal{S} \odot \mathcal{M}_{12}^3$ | $\mathcal{S} \oplus \mathcal{M}_{12}^3$ | $\mathcal{S} \odot \mathcal{M}_{16}^3$ | $\mathcal{S} \oplus \mathcal{M}_{16}^3$ |
|---|---|---|---|---|
| solved | 24159 | 24701 | 25100 | 25397 |
| $\mathcal{S}\%$ | +61.1% | +64.8% | +68.0% | +70.0% |
| $\mathcal{S}+$ | +9761 | +10063 | +10476 | +10647 |
| $\mathcal{S}-$ | -535 | -295 | -309 | -183 |

# Neural Clause Selection in Vampire (M. Suda)

**Deepire: Similar to ENIGMA:**

- build a *classifier* for recognizing *good* clauses
- *good* are those that appeared in past proofs

**Deepire's contributions:**

- Learn from clause *derivation trees only*
  *Not looking at what it says, just who its ancestors were.*
- Integrate using *layered clause queues*
  *A smooth improvement of the base clause selection strategy.*
- Tree Neural Networks: constant work per derived clause
- A signature agnostic approach
- Delayed evaluation trick (not all derived need to be evaluated)

**Preliminary Evaluation on Mizar "57880"**

- Learn from 63595 proofs of 23071 problems (three 30s runs)
- Deepire solves 26217 (i.e. +4054) problems in a *single 10s run*

# TacticToe: mid-level ITP Guidance (Gauthier'17,18)

- TTT learns from human and its own tactical HOL4 proofs
- No translation or reconstruction needed - native tactical proofs
- Fully integrated with HOL4 and easy to use
- Similar to rlCoP: policy/value learning for applying tactics in a state
- However much more technically challenging - a real breakthrough:
  - tactic and goal state recording
  - tactic argument abstraction
  - absolutization of tactic names
  - nontrivial evaluation issues
  - these issues have often more impact than adding better learners
- policy: which tactic/parameters to choose for a current goal?
- value: how likely is this proof state succeed?
- 66% of HOL4 toplevel proofs in 60s (better than a hammer!)
- similar recent work for Isabelle (Nagashima 2018), HOL Light (Google)

# Tactician: Tactical Guidance for Coq (Blaauwbroek'20)

- Tactical guidance of Coq proofs
- Technically very challenging to do right - the Coq internals again nontrivial
- 39.3% on the Coq standard library, 56.7% in a union with CoqHammer (orthogonal)
- Fast approximate hashing for k-NN makes a lot of difference
- Speed more important than better learners
- Fully integrated with Coq, should work for any development
- User friendly, installation friendly, integration friendly and maintenance friendly
- Took several years, but could become a very common tool for Coq formalizers

# Symbolic Rewriting with NNs

- Recurrent NNs with attention good at the inf2formal task
- Piotrowski 2018/19: Experiments with using RNNs for symbolic rewriting
- We can learn rewrite rules from sufficiently many data
- 80-90% success on AIM datasets, 70-99% on normalizing polynomials
- again, complements symbolic methods like ILP that suffer on big data
- in 2019 similar tasks taken up by Facebook - integration, etc.

# Symbolic Rewriting Datasets

Table: Examples in the AIM data set.

| Rewrite rule: | Before rewriting: | After rewriting: |
|---|---|---|
| `b(s(e,v1),e)=v1` | `k(b(s(e,v1),e),v0)` | `k(v1,v0)` |
| `o(V0,e)=V0` | `t(v0,o(v1,o(v2,e)))` | `t(v0,o(v1,v2))` |

Table: Examples in the polynomial data set.

| Before rewriting: | After rewriting: |
|---|---|
| `(x * (x + 1)) + 1` | `x ^ 2 + x + 1` |
| `(2 * y) + 1 + (y * y)` | `y ^ 2 + 2 * y + 1` |
| `(x + 2) * ((2 * x) + 1) + (y + 1)` | `2 * x ^ 2 + 5 * x + y + 3` |

# RL for Normalization and Synthesis Tasks

- Gauthier'19,20:
- Tree Neural Nets and RL (MCTS, policy/value) for:
- Guiding normalization in Robinson arithmetic
- Guiding synthesis of combinators for a given lambda expression
- Guiding synthesis of a diophantine equation characterizing a given set
- Quite encouraging results with a good curriculum (LPAR, CICM)
- Motivated by his TacticToe: argument synthesis and conjecturing is the big missing piece
- Unlike Piotrowski's RNNs/transformers, the results are series of applications of correct/explainable rules
- Gauthier's deep RL framework verifies the whole series (proof) in HOL4

# RL for Normalization and Synthesis Tasks - teaser

- J. Piepenbrock (to be submitted): greatly improved RL for
- Gauthier's normalization in Robinson arithmetic
- Achieved good performance also on the polynomial normalization tasks
- Achieves performance similar to a top equational prover on the AIM problems
- Exciting: again, this is all in the verifiable/explainable proof format

# More on Conjecturing in Mathematics

- Targeted: generate intermediate lemmas (cuts) for a harder conjecture
- Unrestricted (theory exploration):
- Creation of interesting conjectures based on the previous theory
- One of the most interesting activities mathematicians do (how?)
- Higher-level AI/reasoning task - can we learn it?
- If so, we have solved math:
- ... just (recursively) divide Fermat into many subtasks ...
- ... and conquer (I mean: hammer) them away

# A bit of conjecturing history

- The topic goes back at least to Lenat (AM) and Fajtlowicz (Graffiti)
- Combined with automated theorem proving by Colton et al. in early 2000s (HR)
- Theory exploration for Isabelle by Johansson et al (Hipster)
- Several learning-based/neural approaches by our groups since 2015:
- Based mainly on learning analogies and informalization followed by probabilistic/neural disambiguation ...
- ... Gauthier, Kaliszyk, Chvalovsky, Piotrowski, Goertzel, Wang, Brown, JU

# Conjecturing and Proof Synthesis by Neural Language models

- Karpathy'15 - RNN experiments with generating fake Math over Stacks
- I have tried to use that for formal math in 2016 but it looked weak
- GPT (-2,3) looks stronger
- Renewed experiments in 2020 on:
- All Mizar articles, stripped of comments and concatenated together (78M)
- Articles with added context/disambiguation (156M) (types, names, thesis)
- TPTP proofs of 28271 Mizar/MPTP theorems by E/ENIGMA (658M)
- Just the conjecture and premises needed for the 28271 proofs printed in prefix notation
- Quite interesting results, server for Mizar authors
- Quickly taken up by others on HOL, Isabelle, MetaMath ...

# Can you find the flaw(s) in this fake GPT-2 proof?

```
:: generated theorem with "proof"
theorem Th23: :: STIRL2_1:23
for X, Y being finite set st not X is empty & X c= Y
& card X = card Y holds X = Y
proof
  let X, Y be finite set ;
:: thesis: not X is empty & X c= Y & card X = card Y implies X = Y
  assume that
  A1: not X is empty and A2: X c= Y and A3: card X = card Y ;
:: thesis: X = Y
  card (Y \ X) = (card Y) - (card X) by A1, A3, CARD_2:44;
  then A4: card (Y \ X) = ((card Y) - 1) - (card X) by CARD_1:30;
  X = Y \ X by A2, A3, Th22;
  hence X = Y by A4, XBOOLE_0:def_10;
:: thesis: verum
end;
```

Figure: Fake full declarative GPT-2 “proof” - typechecks!

# Mizar autocompletion server in action

Applications Places 3,24GHz Wed 09:07 Wed 09:07

GPT-2 generator trained on Mizar - Chromium

Not secure | grid01.ciirc.cvut.cz:5500

Number of samples (fewer is faster)

3

Temperature (lower is less chaotic)

1.0

Length of output (shorter is faster)

30

Generate

## Sample 1

```
theorem Th0: :: CARD_1:333
for M, N being Cardinal holds card M c= M \/ N
proof
let M, N be Cardinal; ::_thesis: card M c= M \/
```

## Sample 2

```
theorem Th0: :: CARD_1:333
for M, N being Cardinal holds M *` N is Cardinal
proof
let M, N be Cardinal; ::_thesis: M *` N is Cardinal
cf (
```

## Sample 3

```
theorem Th0: :: CARD_1:333
for M, N being Cardinal holds Sum (M --> N) c= M *` N
proof
let M, N be Cardinal; ::_thesis: Sum (M
```

[github]

Figure: MGG - Mizar Gibberish Generator.

# Proving the conditioned completions - MizAR hammer

Applications Places 4,81 GHz Wed 14:42 Wed 14:42

emacs@dell

File Edit Options Buffers Tools Index Mizar Hide/Show Help

Save Undo

```
begin

for M, N being Cardinal holds card M c= M \/ N by XBOOLE_1:7,CARD_3:44,CARD_1:7,CARD_1:3; :: [ATP details]

for X, Y being finite set st not X is empty & X c= Y & card X = card Y holds X = Y by CARD_FIN:1; :: [ATP details]

for M, N being Cardinal holds
( M in N iff card M c= N ) by Unsolved; :: [ATP details]

for M, N being Cardinal holds
( M in N iff card M in N ) by CARD_3:44,CARD_1:9; :: [ATP details]

for M, N being Cardinal holds Sum (M --> N) = M *` N by CARD_2:65; :: [ATP details]


for M, N being Cardinal holds M /\ (union N) in N by Unsolved; :: [ATP details]


for M, N being Cardinal holds M *` N = N *` M by ATP-Unsolved; :: [ATP details]
```

-:--- **card_tst.miz** 3% L47 (Mizar Errors:2 hs Undo-Tree)

Wrote /home/urban/mizwrk/7.13.01_4.181.1147/tst8/card_tst.miz

# A correct conjecture that was too hard to prove

- Kinyon and Stanovsky (algebraists) confirmed that this cut is valid:

```
theorem Th10: :: GROUPP_1:10
for G being finite Group for N being normal Subgroup of G st
N is Subgroup of center G & G ./. N is cyclic holds G is commutative

The generalization that avoids finiteness:

for G being Group for N being normal Subgroup of G st
N is Subgroup of center G & G ./. N is cyclic holds G is commutative
```

# Gibberish Generator Provoking Algebraists

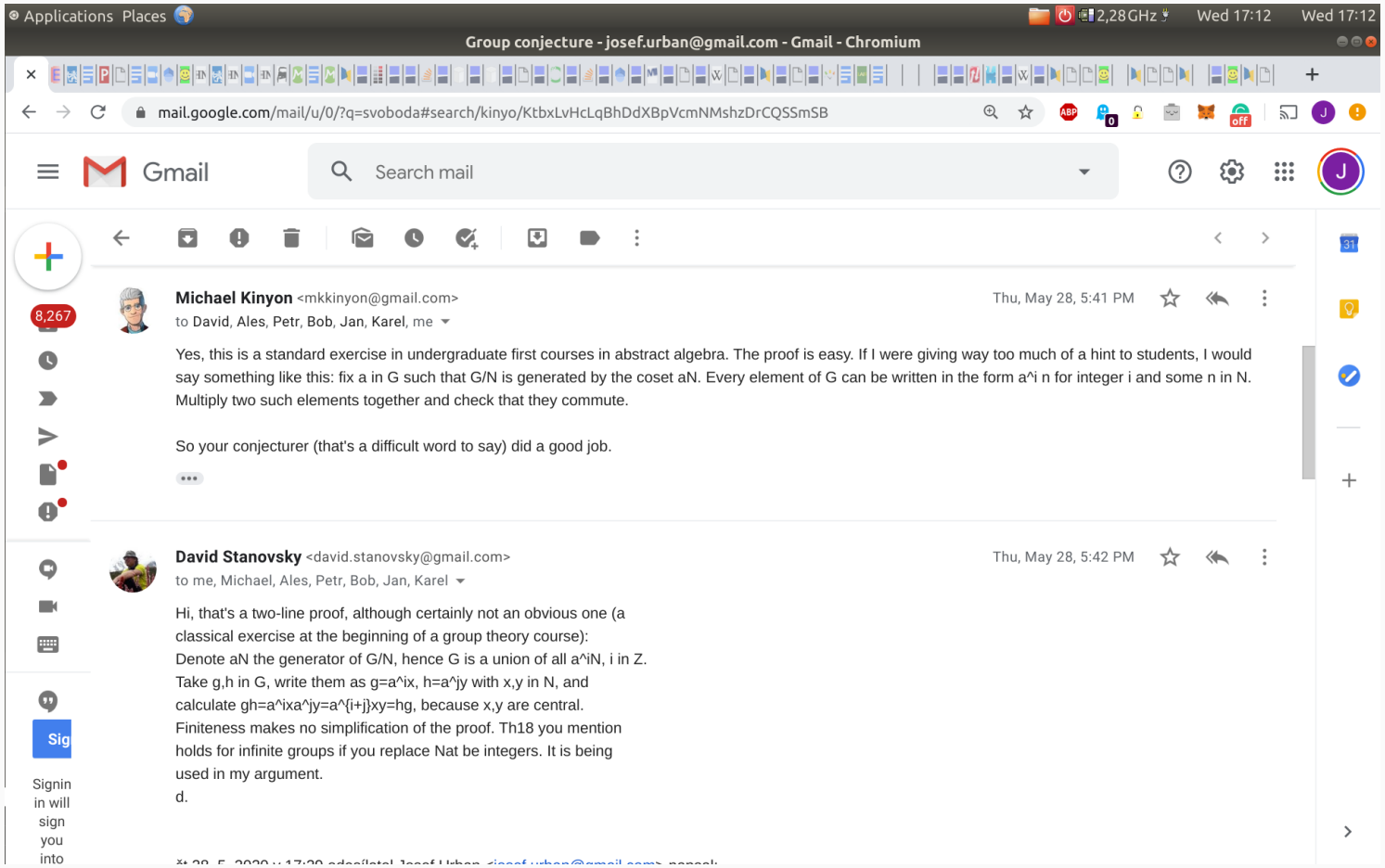



Figure: First successes in making mathematicians comment on AI.

# More cuts

- In total 33100 in this experiment
- Ca 9k proved by trained ENIGMA
- Some are clearly false, yet quite natural to ask:

```
theorem :: SINCOS10:17
sec is increasing on [0, pi/2)

leads to conjecturing the following:

Every differentiable function is increasing.
```

# Neural Autoformalization (Wang et al., 2018)

- generate ca 1M Latex/Mizar pairs based on Bancerek's work
- train neural seq-to-seq translation models (Luong – NMT)
- evaluate on about 100k examples
- many architectures tested, some work much better than others
- very important latest invention: *attention* in the seq-to-seq models
- more data very important for neural training – our biggest bottleneck (you can help!)
- Recent addition: unsupervised methods (Lample et all 2018) – no need for aligned data!

# Neural Autoformalization data

| | |
|---|---|
| Rendered LaTeX | If $X \subseteq Y \subseteq Z$, then $X \subseteq Z$. |
| Mizar | `X c= Y & Y c= Z implies X c= Z;` |
| Tokenized Mizar | `X c= Y & Y c= Z implies X c= Z ;` |
| LaTeX | `If $X \subseteq Y \subseteq Z$, then $X \subseteq Z$.` |
| Tokenized LaTeX | `If $ X \subseteq Y \subseteq Z $ , then $ X \subseteq Z $ .` |

# Neural Autoformalization results

| Parameter | Final Test Perplexity | Final Test BLEU | Identical Statements (%) | Identical No-overlap (%) |
|---|---|---|---|---|
| 128 Units | 3.06 | 41.1 | 40121 (38.12%) | 6458 (13.43%) |
| 256 Units | 1.59 | 64.2 | 63433 (60.27%) | 19685 (40.92%) |
| 512 Units | 1.6 | **67.9** | 66361 (63.05%) | 21506 (44.71%) |
| 1024 Units | **1.51** | 61.6 | **69179 (65.73%)** | **22978 (47.77%)** |
| 2048 Units | 2.02 | 60 | 59637 (56.66%) | 16284 (33.85%) |

# Neural Fun – Performance after Some Training

| | |
|---|---|
| Rendered LaTeX | Suppose $s_8$ is convergent and $s_7$ is convergent . Then $\mathop{\rm lim}(s_8+s_7) = \mathop{\rm lim}\, s_8+\mathop{\rm lim}\, s_7$ |
| Input LaTeX | `Suppose $ { s _ { 8 } } $ is convergent and $ { s _ { 7 } } $ is convergent . Then $ \mathop { \rm lim } ( { s _ { 8 } } { + } { s _ { 7 } } ) \mathrel { = } \mathop { \rm lim } { s _ { 8 } } { + } \mathop { \rm lim } { s _ { 7 } } $ .` |
| Correct | `seq1 is convergent & seq2 is convergent implies lim ( seq1 + seq2 ) = ( lim seq1 ) + ( lim seq2 ) ;` |
| Snapshot-1000 | `x in dom f implies ( x * y ) * ( f \| ( x \| ( y \| ( y \| y ) ) ) ) = ( x \| ( y \| ( y \| ( y \| y ) ) ) ) ) ;` |
| Snapshot-2000 | `seq is summable implies seq is summable ;` |
| Snapshot-3000 | `seq is convergent & lim seq = 0c implies seq = seq ;` |
| Snapshot-4000 | `seq is convergent & lim seq = lim seq implies seq1 + seq2 is convergent ;` |
| Snapshot-5000 | `seq1 is convergent & lim seq2 = lim seq2 implies lim_inf seq1 = lim_inf seq2 ;` |
| Snapshot-6000 | `seq is convergent & lim seq = lim seq implies seq1 + seq2 is convergent ;` |
| Snapshot-7000 | `seq is convergent & seq9 is convergent implies lim ( seq + seq9 ) = ( lim seq ) + ( lim seq9 ) ;` |

# Unsupervised NMT Fun on Short Formulas

```
len <* a *> = 1 ;
assume i < len q ;
len <* q *> = 1 ;
s = apply ( v2 , v1 ast t ) ;
s . ( i + 1 ) = tt . ( i + 1 )
1 + j <= len v2 ;
1 + j + 0 <= len v2 + 1 ;
let i be Nat ;
assume v is_applicable_to t ;
let t be type of T ;
a ast t in downarrow t ;
t9 in types a ;
a ast t <= t ;
A is_applicable_to t ;
Carrier ( f ) c= B
u in B or u in { v } ;
F . w in w & F . w in I ;
GG . y in rng HH ;
a * L = Z_ZeroLC ( V ) ;
not u in { v } ;
u <> v ;
v - w = v1 - w1 ;
v + w = v1 + w1 ;
x in A & y in A ;
```

```
len <* a *> = 1 ;
i < len q ;
len <* q *> = 1 ;
s = apply ( v2 , v1 ) . t ;
s . ( i + 1 ) = tau1 . ( i + 1 )
1 + j <= len v2 ;
1 + j + 0 <= len v2 + 1 ;
i is_at_least_length_of p ;
not v is applicable ;
t is_orientedpath_of v1 , v2 , T ;
a *' in downarrow t ;
t `2 in types a ;
a *' <= t ;
A is applicable ;
support ppf n c= B
u in B or u in { v } ;
F . w in F & F . w in I ;
G0 . y in rng ( H1 ./. y ) ;
a * L = ZeroLC ( V ) ;
u >> v ;
u <> v ;
vw = v1 - w1 ;
v + w = v1 + w1 ;
assume [ x , y ] in A ;
```

# Acknowledgments


- Prague Automated Reasoning Group `http://arg.ciirc.cvut.cz/`:
  - Jan Jakubuv, Chad Brown, Martin Suda, Karel Chvalovsky, Bob Veroff, Zar Goertzel, Bartosz Piotrowski, Lasse Blaauwbroek, Martin Smolik, Jiri Vyskocil, Petr Pudlak, David Stanovsky, Krystof Hoder, ...
- HOL(y)Hammer group in Innsbruck:
  - Cezary Kaliszyk, Thibault Gauthier, Michael Faerber, Yutaka Nagashima, Shawn Wang
- ATP and ITP people:
  - Stephan Schulz, Geoff Sutcliffe, Andrej Voronkov, Kostya Korovin, Larry Paulson, Jasmin Blanchette, John Harrison, Tom Hales, Tobias Nipkow, Andrzej Trybulec, Piotr Rudnicki, Adam Pease, ...
- Learning2Reason people at Radboud University Nijmegen:
  - Herman Geuvers, Tom Heskes, Daniel Kuehlwein, Evgeni Tsivtsivadze, ....
- Google Research: Christian Szegedy, Geoffrey Irving, Alex Alemi, Francois Chollet, Sarah Loos
- ... and many more ...
- Funding: Marie-Curie, NWO, ERC

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# Thanks and Advertisement

- Thanks for your attention!
- AITP – Artificial Intelligence and Theorem Proving
- September 5–10, 2021, Aussois, France, `aitp-conference.org`
- ATP/ITP/Math vs AI/Machine-Learning people, Computational linguists
- Discussion-oriented and experimental - submit a talk abstract!
- Grown to 80 people in 2019
- Will be hybrid in 2021 as in 2020